

Pathology-Level ML Model Performance Summary (Test Set)
Classical ML models trained on neural network pathology predictions

Neural Network Model	Pathology	ML Algorithm	AUC Mean	AUC 95% CI	Samples	Positive Rate	Cross-Val Folds
Attention Pool	A Lines	Random Forest	0.687	[0.604-0.769]	43689	0.472	5
Attention Pool	A Lines	Logistic Regression	0.742	[0.673-0.810]	43689	0.472	5
Attention Pool	A Lines	XGBoost	0.727	[0.656-0.799]	43689	0.472	5
Attention Pool	Large Consolidation	Random Forest	0.784	[0.668-0.901]	43689	0.110	5
Attention Pool	Large Consolidation	Logistic Regression	0.875	[0.817-0.934]	43689	0.110	5
Attention Pool	Large Consolidation	XGBoost	0.828	[0.725-0.931]	43689	0.110	5
Attention Pool	Pleural Effusion	Random Forest	0.675	[0.651-0.698]	43689	0.011	5
Attention Pool	Pleural Effusion	Logistic Regression	0.739	[0.715-0.763]	43689	0.011	5
Attention Pool	Pleural Effusion	XGBoost	0.821	[0.790-0.853]	43689	0.011	5
Attention Pool	Other Pathology	Random Forest	0.590	[0.534-0.647]	43689	0.405	5
Attention Pool	Other Pathology	Logistic Regression	0.630	[0.617-0.643]	43689	0.405	5
Attention Pool	Other Pathology	XGBoost	0.627	[0.570-0.683]	43689	0.405	5
Original	A Lines	Random Forest	0.704	[0.660-0.749]	43689	0.472	5
Original	A Lines	Logistic Regression	0.761	[0.725-0.796]	43689	0.472	5
Original	A Lines	XGBoost	0.745	[0.702-0.788]	43689	0.472	5
Original	Large Consolidation	Random Forest	0.812	[0.760-0.865]	43689	0.110	5
Original	Large Consolidation	Logistic Regression	0.899	[0.877-0.922]	43689	0.110	5
Original	Large Consolidation	XGBoost	0.868	[0.856-0.881]	43689	0.110	5
Original	Pleural Effusion	Random Forest	0.663	[0.596-0.730]	43689	0.011	5
Original	Pleural Effusion	Logistic Regression	0.846	[0.817-0.876]	43689	0.011	5
Original	Pleural Effusion	XGBoost	0.834	[0.785-0.883]	43689	0.011	5
Original	Other Pathology	Random Forest	0.617	[0.604-0.630]	43689	0.405	5
Original	Other Pathology	Logistic Regression	0.647	[0.625-0.669]	43689	0.405	5
Original	Other Pathology	XGBoost	0.646	[0.635-0.658]	43689	0.405	5
Original Noinitweights	A Lines	Random Forest	0.702	[0.610-0.793]	43689	0.472	5
Original Noinitweights	A Lines	Logistic Regression	0.753	[0.688-0.817]	43689	0.472	5
Original Noinitweights	A Lines	XGBoost	0.737	[0.651-0.823]	43689	0.472	5
Original Noinitweights	Large Consolidation	Random Forest	0.747	[0.635-0.858]	43689	0.110	5
Original Noinitweights	Large Consolidation	Logistic Regression	0.867	[0.826-0.908]	43689	0.110	5
Original Noinitweights	Large Consolidation	XGBoost	0.806	[0.730-0.882]	43689	0.110	5
Original Noinitweights	Pleural Effusion	Random Forest	0.613	[0.552-0.674]	43689	0.011	5
Original Noinitweights	Pleural Effusion	Logistic Regression	0.843	[0.782-0.903]	43689	0.011	5
Original Noinitweights	Pleural Effusion	XGBoost	0.785	[0.728-0.842]	43689	0.011	5
Original Noinitweights	Other Pathology	Random Forest	0.601	[0.561-0.640]	43689	0.405	5
Original Noinitweights	Other Pathology	Logistic Regression	0.653	[0.609-0.697]	43689	0.405	5
Original Noinitweights	Other Pathology	XGBoost	0.640	[0.604-0.676]	43689	0.405	5